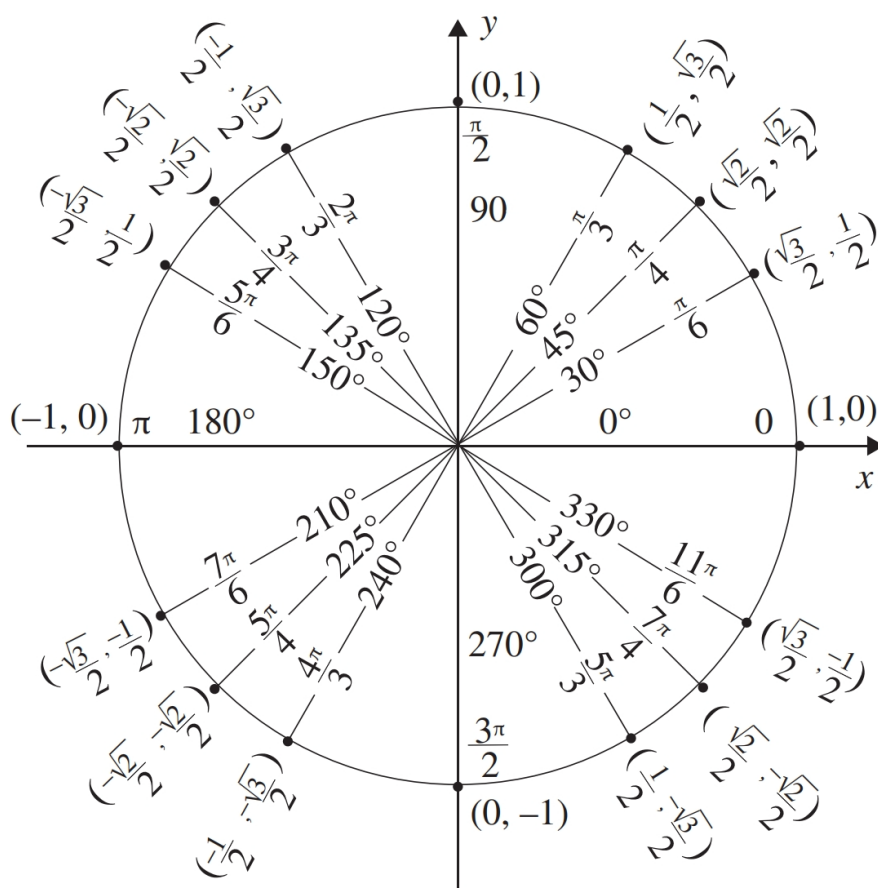


Inverse Trigonometric Functions

Section 3.5

A. Unit Circle and Common Values



Degree	Radians	Sin θ	cos θ	tan θ	cot θ	sec θ	csc θ
0°	0	0°	1	0°	Undefined	1	Undefined
30°	$\pi/6$	$1/2$	$\sqrt{3}/2$	$\sqrt{3}/3$	$\sqrt{3}$	$2\sqrt{3}/3$	2
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1	1	$\sqrt{2}$	$\sqrt{2}$
60°	$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$	$\sqrt{3}/3$	2	$2\sqrt{3}/3$
90°	$\pi/2$	1	0	Undefined	0	Undefined	1
120°	$2\pi/3$	$\sqrt{3}/2$	$-1/2$	$-\sqrt{3}$	$-\sqrt{3}/3$	-2	$2\sqrt{3}/3$
135°	$3\pi/4$	$\sqrt{2}/2$	$-\sqrt{2}/2$	-1	-1	$-\sqrt{2}$	$\sqrt{2}$
150°	$5\pi/6$	$1/2$	$-\sqrt{3}/2$	$-\sqrt{3}/3$	$-\sqrt{3}$	$-2\sqrt{3}/3$	2
180°	π	0	-1	0	Undefined	-1	Undefined
210°	$7\pi/6$	$-1/2$	$-\sqrt{3}/2$	$\sqrt{3}/3$	$\sqrt{3}$	$-2\sqrt{3}/3$	-2
225°	$5\pi/4$	$-\sqrt{2}/2$	$-\sqrt{2}/2$	1	1	$-\sqrt{2}$	$-\sqrt{2}$
240°	$4\pi/3$	$-\sqrt{3}/2$	$-1/2$	$\sqrt{3}$	$\sqrt{3}/3$	-2	$-2\sqrt{3}/3$
270°	$3\pi/2$	-1	0	Undefined	0	Undefined	-1
300°	$5\pi/2$	$-\sqrt{3}/2$	$1/2$	$-\sqrt{3}$	$-\sqrt{3}$	2	$-2\sqrt{3}/3$
315°	$7\pi/4$	$-\sqrt{2}/2$	$\sqrt{2}/2$	-1	-1	$\sqrt{2}$	$-\sqrt{2}$
330°	$11\pi/6$	$-1/2$	$\sqrt{3}/2$	$-\sqrt{3}/3$	$-\sqrt{3}$	$2\sqrt{3}/3$	-2
360°	2π	0	1	0	Undefined	1	Undefined

Examples

1. (Video) Give the exact value of $\cos^{-1}\left(\frac{1}{2}\right)$

2. (Video) Give the exact value of $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$

3. **(Video)** Give the exact value of $\sin^{-1}\left(\frac{1}{2}\right)$

4. **(Video)** Give the exact value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

5. **(Video)** Give the exact value of $\tan^{-1}(\sqrt{3})$

6. **(Video)** Give the exact value of $\tan^{-1}(1)$

7. **(Video)** Find $\lim_{x \rightarrow \infty} \tan^{-1} x$

8. **(Video)** Find $\lim_{x \rightarrow -\infty} \tan^{-1} x$

B. Derivatives of Inverse Trigonometric Functions (You Must Know These!)

1. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

2. $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$

3. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

4. $\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$

5. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$

6. $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$

Examples

9. (video) Find $\frac{d}{dx}(5\sec^{-1}x + 2x^2 + 3)$

10. (video) Find $\frac{d}{dx}(5\cos^{-1}(3x^2 + 5))$

11. (video) Find $\frac{d}{dx}(\tan^{-1}(5x^2 + 17))$

12. Let $f(x) = (\tan^{-1} x)^7$. Find $f'(x)$.

13. Let $f(x) = \cos^{-1}(e^{8x})$. Find $f'(x)$.

14. Find the limit: $\lim_{x \rightarrow 1^-} \sin^{-1} x$

15. Find the limit: $\lim_{x \rightarrow \infty} \arctan(-e^x)$